Tree Height Triangulation Shootout

Measure a tree without climbing it

At a glance: Measure a tree without climbing it. Plan about 60 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

By the end of this activity, each student can:

- Explain the core science of this challenge: angles, distance, estimation, remote sensing (Understand).
- Design a fair test by controlling one variable while holding others constant (Apply).
- Use their own data to decide whether a redesign improved the result (Analyze).
- Defend a final claim with specific evidence (Evaluate).

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- 4 paper clinometers (plus 2 spares; GLOBE-style template on cardstock, with a straw and a weighted string)
- String and washers, shared (1 roll string, 1 pack washers; the plumb line for each clinometer)
- 4 twelve-inch rulers (plus 2 spares; to measure each student's eye height)
- 1 long open-reel tape measure (100 ft / 30 m) to pre-mark a fixed standoff distance from the reference tree; the 12 in rulers cannot set this distance
- 4 clipboards (plus 1 spare)
- pencils and a calculator or printed tan table per team, shared

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

03 No PPE: this is an outdoor measurement activity with no chemicals, water, or soil. Set up sun and hydration support, run the weather/heat-index check, confirm the buddy system and boundaries, and pre-mark the tree-height standoff distance with the long tape.

04 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Measure a tree without climbing it." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: approved materials only, change one variable at a time, record at least one data point before any redesign, and follow the safety notes.

Phase 4 Build or solve

0:18 - 0:38

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Sight and compute

From the marked standoff distance, each team sights the treetop through the straw, reads and averages the plumb-string angle, measures eye height with the 12 in ruler, and computes tree height = eye height + (distance x tan(angle)).

Phase 6 Re-measure to improve

Each team takes a second set of sightings (level the eye, average more readings, recheck eye height) and reports whether their estimate moved closer to the reference.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, and reset the station. Return the clinometers, string, washers, rulers, the tape, and clipboards to the bin; there is no PPE to remove and nothing to wash.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A team's height estimate is far from the reference. Fix: Check the clinometer reads 0 when level and 90 straight up, confirm the team sighted the very top from the marked distance (not a guessed distance), average several readings, and confirm they added eye height.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to angles give height.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- What was your biggest source of error?
- Why does eye height matter in the calculation?
- Where is remote height measurement used in the field?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Accuracy	50
Method clarity	20
Speed	20
Teamwork	10
Total	**100**

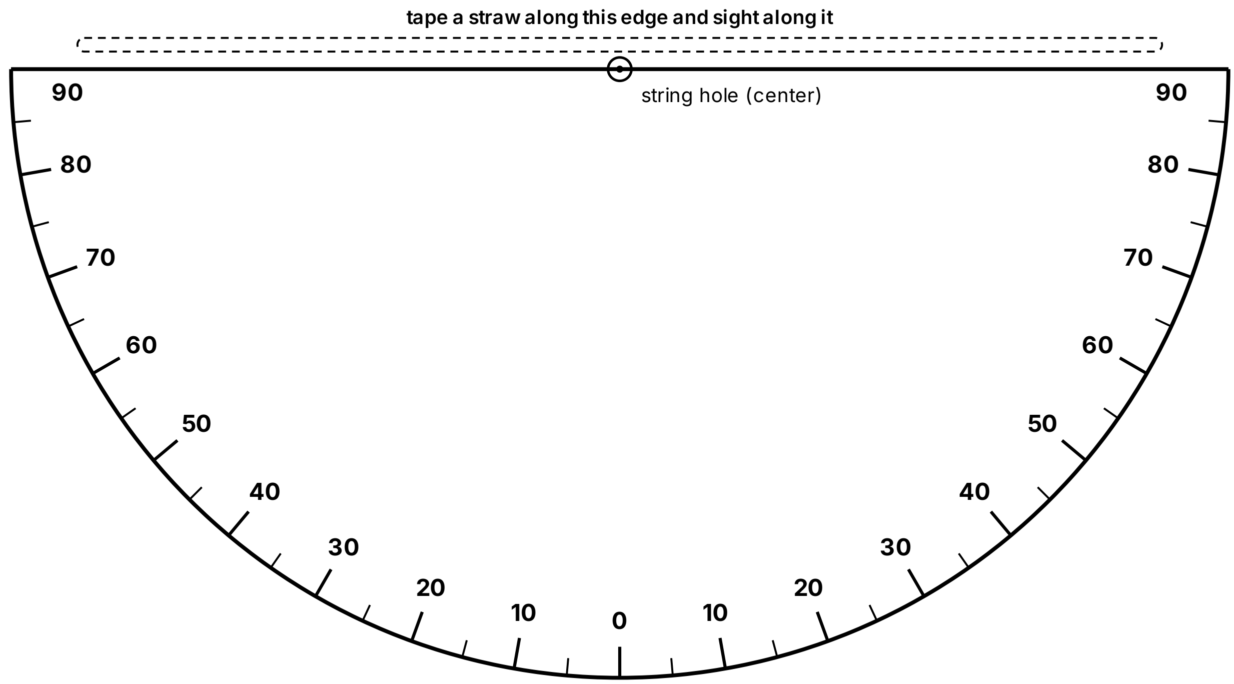
CLEANUP AND SOURCES

Return the clinometers, string, washers, rulers, the tape, and clipboards to the bin and collect score slips. The only consumable is the printed clinometer template; there are no surfaces to wipe.

SOURCES GLOBE Trees Activities

Paper clinometer and tan table

Print on cardstock: 4 per station plus 2 spares. This is the one consumable the activity needs. Read the angle to the treetop, then use the tan table (no calculator needed) to find the height.



Cut along the flat top edge and the curved outline.

ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer on the end as a weight.

04 Sight the treetop through the straw, let the string hang, then pinch it against the scale and read the angle.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor pre-marked with the long tape.

ANGLE	TAN
15°	0.27
20°	0.36
25°	0.47
30°	0.58
35°	0.70
40°	0.84
45°	1.00

ANGLE	TAN
50°	1.19
55°	1.43
60°	1.73
65°	2.14
70°	2.75
75°	3.73

Example: eye height 1.5 m, distance 15 m, angle 40° gives 1.5 + 15 × 0.84 = about 14.1 m.